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Tables → x

chairs → y

Profit $= 250x + 75y$

$2500x + 500y \le 50{,}000$

$x + y \le 60$

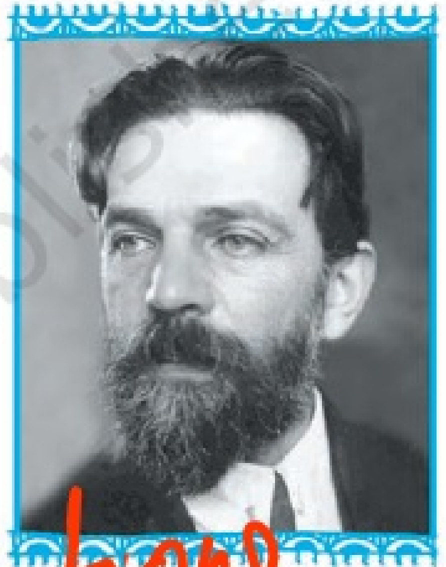
$x \ge 0$

$y \ge 0$

$$\boxed{Z = 250x + 75y}$$

maximise / minimise

12.1 Introduction



L. Kantorovich

In earlier classes, we have discussed systems of linear equations and their applications in day to day problems. In Class XI, we have studied linear inequalities and systems of linear inequalities in two variables and their solutions by graphical method. Many applications in mathematics involve systems of inequalities/equations. In this chapter, we shall apply the systems of linear inequalities/equations to solve some real life problems of the type as given below:

A furniture dealer deals in only two items–tables and chairs. He has Rs 50,000 to invest and has storage space of at most 60 pieces. A table costs Rs 2500 and a chair Rs 500. He estimates that from the sale of one table, he can make a profit of Rs 250 and that from the sale of one chair a profit of Rs 75. He wants to know how many tables and chairs he should buy from the available money so as to maximise his total profit, assuming that he can sell all the items which he buys.

linear inequations or linear constraints

Such type of problems which seek to maximise (or, minimise) profit (or, cost) form a general class of problems called **optimisation problems**. Thus, an optimisation problem may involve finding maximum profit, minimum cost, or minimum use of resources etc.

A special but a very important class of optimisation problems is **linear programming problem.** The above stated optimisation problem is an example of linear programming problem. Linear programming problems are of much interest because of their wide applicability in industry, commerce, management science etc.

objective functions

In this chapter, we shall study some linear programming problems and their solutions by graphical method only, though there are many other methods also to solve such problems.

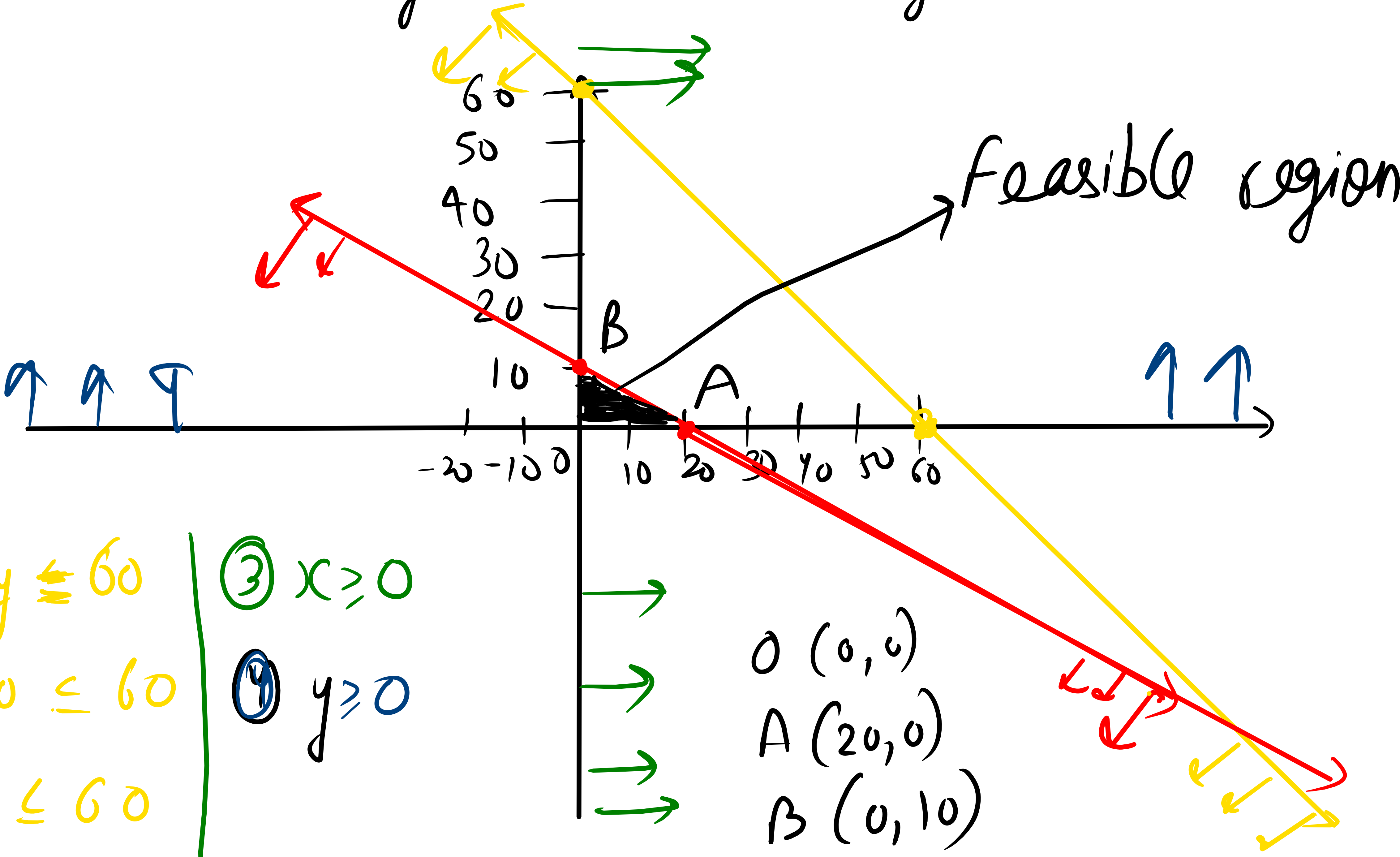
$$2500x + 5000y \le 50{,}000 \Rightarrow 25x + 50y \le 500 \Rightarrow x + 2y \le 20$$

$$x + y \le 60$$

$$x \geq 0$$

$$y \geq 0$$

$$\boxed{Z = 250x + 75y}$$



① $x + 2y = 20$

$x + 2y \le 20$

$0 + 0 \le 20$

$0 \le 20$

② $x + y \le 60$

$0 + 0 \le 60$

$0 \le 60$

③ $x \geq 0$

④ $y \geq 0$

Points	$z = 250x + 75y$
$(0, 0)$	$z = 0$
$A(20, 0)$	$z = 5000$
$B(0, 10)$	$z = 750$

→ maximum

maximum Profit
Comes at point $A(20, 0)$

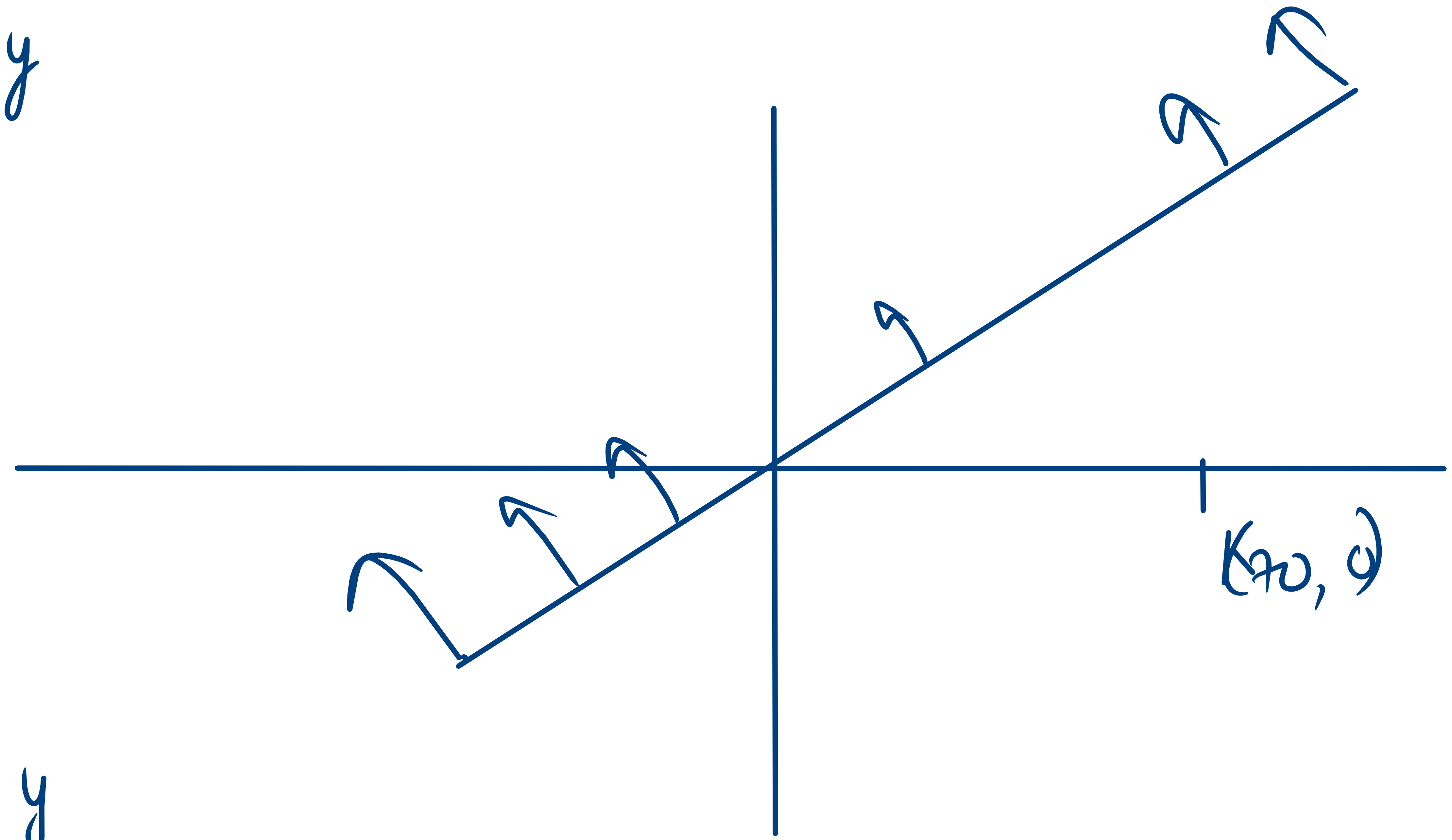
$x = 20$
$y = 0$

max. Profit = 5,000

Tables = 20
chairs = 0

$x \le y$

$y = x$



① $x \le y$

$70 \le 0$

NO

has no minimum value subject to the given constraints.

In the above example, can you say whether $z = -50x + 20y$ has the maximum value 100 at (0,5)? For this, check whether the graph of $-50x + 20y > 100$ has points in common with the feasible region. (Why?)

Example 5 Minimise $Z = 3x + 2y$

subject to the constraints:

$$x + y \geq 8 \quad \ldots (1)$$

$$3x + 5y \leq 15 \quad \ldots (2)$$

$$x \geq 0,\ y \geq 0 \quad \ldots (3)$$

Solution Let us graph the inequalities (1) to (3) (Fig 12.6). Is there any feasible region? Why is so?

From Fig 12.6, you can see that there is no point satisfying all the constraints simultaneously. Thus, the problem is having no feasible region and hence no feasible solution.

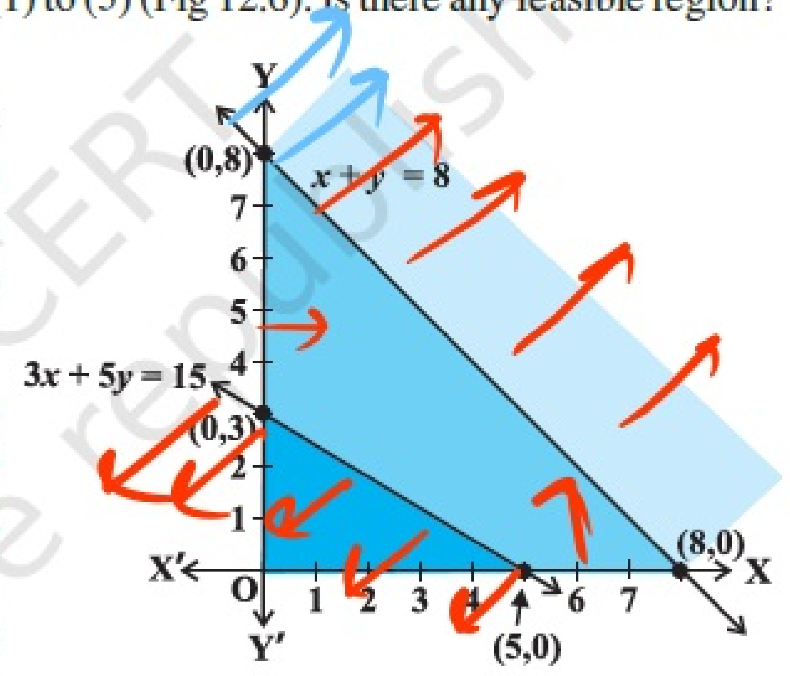


Fig 12.6

Remarks From the examples which we have discussed so far, we notice some general features of linear programming problems:

(i) The feasible region is always a convex region.

(ii) The maximum (or minimum) solution of the objective function occurs at the vertex (corner) of the feasible region. If two corner points produce the same maximum (or minimum) value of the objective function, then every point on the line segment joining these

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Example 4 Determine graphically the minimum value of the objective function

$$Z = -50x + 20y \qquad \text{... (1)}$$

subject to the constraints:

$$2x - y \geq -5 \qquad \text{... (2)}$$
$$3x + y \geq 3 \qquad \text{... (3)}$$
$$2x - 3y \leq 12 \qquad \text{... (4)}$$
$$x \geq 0,\ y \geq 0 \qquad \text{... (5)}$$

① $2x - y = -5$

$y = 1$

$x = -2$

$x = 0 ;\ y = 5$

$0 - 0 \geq -5$

$0 \geq -5$

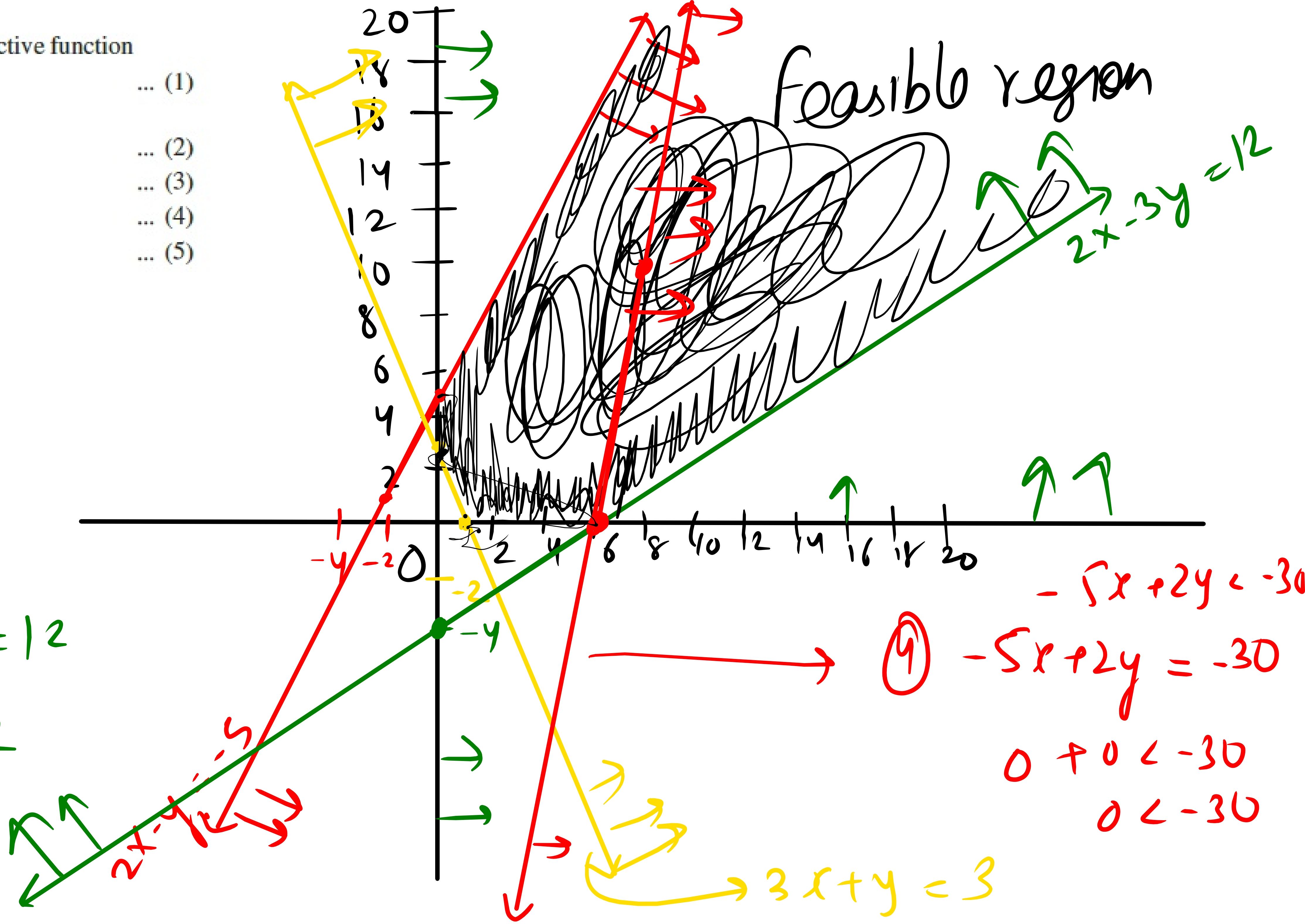
② $3x + y = 3$

$0 \geq 3$

Not

③ $2x - 3y = 12$

$0 - 0 \leq 12$



① Feasible region → bounded → Put corner points in z, and get max/min value

② unbounded feasible region : $z = ax + by$

max (M)   min (m)

for max value $ax + by \geq M$

for min value $ax + by < m$

After graph, if this new equation has no common point with the feasible region then M, m → will be max/min value otherwise, it does not exist

Corner points	$Z = -50x + 20y$
$(0, 5)$	100
$(0, 3)$	-60
$(1, 0)$	-50
$(6, 0)$	-300 → smallest

$-50x + 20y < -300$

$$\boxed{-5x + 2y < -30}$$ ✓

Graph of $-5x + 2y < -30$ has common points with the feasible region. Hence there is no minimum value.

H.W

Ex-12.1 → Q1, Q2, Q6, Q9, Q10

+

Example - 3 and 4